

Outlier Detection

An online retailer called Buy-a-lot¹ has recently opened in South Africa and wants to know which of their products are hot/trending.

You are given a .csv file of sales records from 31/12/2019 to 29/06/2020, as well as a .csv file of baseline forecasts², which are the total sales per product over the next four weeks. By generating your own forecasts, you are required to determine which of these products, in terms of total volume sold, are expected to increase by 100% or more above the baseline forecasts for the following four weeks. Based on your predictions, these high growth products will be labelled on the company's website as trending products.

Your solution should consist of the following components:

1. Ingestion
 - Read from the .csv files
 - Extract the relevant information from the datasets
 - Note: negative volume orders are returns and should be included. Zero price orders are discounts (e.g. buy one, get one free) and should not be excluded.
 - Note: missing days should be assumed to have zero volume.
2. Modelling
 - Use an applicable statistics/machine learning method (e.g. ARIMA, ETS, GLMs, neural nets, ...) to make 100 forecasts that are the total quantity sold of each product over the following four weeks.
 - If your forecasted volume is greater than the baseline forecast by 100% or more for a given product, this product should be labelled as a 'trending product'.
 - The baseline predictions are given as a total over the following four weeks, so your predictions should also be made as totals over the following four weeks.
3. Simple reporting
 - Write your results to a .csv file with the following column headings: `product_code` and `is_trending`. The `is_trending` column must have either a `TRUE` or `FALSE` value in each row.
 - The company also wants advice about how their system could be improved in the future. In a .txt file, write a short paragraph another way you can think of for detecting trending products and how it could improve on the current method.

The above steps should run end-to-end (.csv to results) with a single call to a script. Your solution must be written using only open source tools (e.g. python, R, ...) so we can run your solution and check your results. If necessary, in a `README` file, specify any external libraries or packages that you used.

¹ A fictitious retailer

² The baseline forecast is based on the mean sales over the last 6 months for a given product